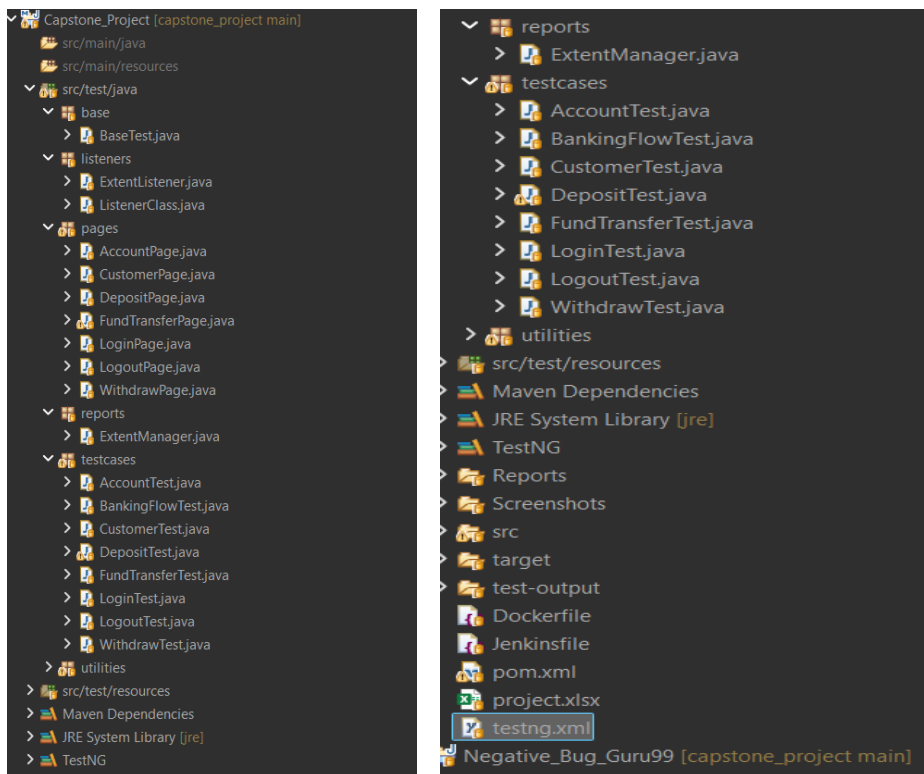


Project Name --- Guru 99 Bank**Summary: -**

The **Guru99 Bank Automation Framework** is a web automation testing project developed using **Java, Selenium WebDriver, TestNG, Maven, Jenkins, docker, Extent Reports, and GitHub** following the **Page Object Model (POM)** design pattern. The framework automates key banking functionalities such as Login Validation, Customer Creation, Account Creation, Deposit, Withdrawal, and Fund Transfer, along with negative test scenarios. It includes reusable page classes, data-driven testing using Excel (Apache POI), screenshot capture on failure, retry mechanisms, TestNG listeners, and detailed Extent Reports. The project is integrated with Jenkins for Continuous Integration, Docker and GitHub for version control. A total of **13 automated test cases** were executed successfully with a **100% pass rate**, and several application defects were identified, documented, and tracked through Jira, demonstrating a complete end-to-end automation testing lifecycle.

Framework of My Project:-

Now the code of all file of my Project:-

Package Name – base

Class Name- BaseTest.Java :-

package base;

import java.time.Duration;

import org.openqa.selenium.WebDriver;

import org.openqa.selenium.chrome.ChromeDriver;

import org.openqa.selenium.chrome.ChromeOptions;

import org.openqa.selenium.edge.EdgeDriver;

import org.openqa.selenium.edge.EdgeOptions;

import org.openqa.selenium.firefox.FirefoxDriver;

import org.openqa.selenium.firefox.FirefoxOptions;

import org.testng.annotations.AfterMethod;

import org.testng.annotations.BeforeMethod;

import org.testng.asserts.SoftAssert;

import pages.LoginPage;

import utilities.ConfigReader;

import utilities.ExcelUtils;

public class BaseTest {

public static WebDriver driver;

public SoftAssert soft;

ConfigReader config;

@BeforeMethod

public void setup() throws Exception {

soft = new SoftAssert();

config = new ConfigReader();

String browser = config.getBrowser();

if(browser.equalsIgnoreCase("chrome")) {

ChromeOptions options = new ChromeOptions();

options.addArguments("--headless=new");

options.addArguments("--no-sandbox");

options.addArguments("--disable-dev-shm-usage");

options.addArguments("--window-size=1920,1080");

driver = new ChromeDriver(options);

System.out.println("Chrome browser started");

} else if(browser.equalsIgnoreCase("edge")) {

EdgeOptions options = new EdgeOptions();

options.addArguments("--headless=new");

driver = new EdgeDriver(options);

```
        System.out.println("Edge Headless Started");

    } else if(browser.equalsIgnoreCase("firefox")) {

        FirefoxOptions options =new FirefoxOptions();

        options.addArguments("--headless");

        driver = new FirefoxDriver(options);

        System.out.println("Firefox Headless Started");
    }

    driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));

    driver.get(config.getUrl());
}

public void login() throws Exception {

    String user =ExcelUtils.getData("Sheet1", 1, 0);
    String pass =ExcelUtils.getData("Sheet1", 1, 1);

    LoginPage lp =new LoginPage(driver);

    lp.enterUser(user);
    lp.enterPass(pass);
    lp.clickLogin();
}
```

@AfterMethod

public void tearDown() {

if(driver != null) {

driver.quit();

}

}

}

Package Name – listeners

Class Name- ExtentListener.java

package listeners;

import org.testng.*;

import com.aventstack.extentreports.*;

import reports.ExtentManager;

public class ExtentListener

implements ITestListener {

ExtentReports extent =

ExtentManager.getReport();

ExtentTest test;

```

public void onTestStart(
    ITestResult result) {

    test = extent.createTest(
        result.getName());
}

public void onTestSuccess(
    ITestResult result) {

    test.pass("Test Passed");
}

public void onTestFailure(
    ITestResult result) {

    test.fail(result.getThrowable());
}

public void onFinish(
    ITestContext context) {

    extent.flush();
}
}

```

Package Name – listeners

Class Name- listenerClass.java

```

package listeners;

```

```
import java.io.IOException;
```

```
import org.testng.ITestListener;
```

```
import org.testng.ITestResult;
```

```
import base.BaseTest;
```

```
import utilities.ScreenshotUtil;
```

```
public class ListenerClass
```

```
implements ITestListener {
```

```
    @Override
```

```
    public void onTestStart(
```

```
        ITestResult result) {
```

```
        System.out.println("Test Started : " + result.getName());
```

```
    }
```

```
    @Override
```

```
    public void onTestSuccess(
```

```
        ITestResult result) {
```

```
        System.out.println("Test Passed : " + result.getName());
```

```
    }
```

```
    @Override
```

```
    public void onTestFailure(
```

```
        ITestResult result) {
```

```
        System.out.println("Test Failed : " + result.getName());
```

```

try {
    ScreenshotUtil.captureScreenshot(BaseTest.driver,result.getName());
} catch (IOException e) {

    e.printStackTrace();
}
}
}

```

Package Name – pages

Class Name- AccountPage.java

```
package pages;
```

```
import org.openqa.selenium.By;
```

```
import org.openqa.selenium.WebDriver;
```

```
import org.openqa.selenium.support.ui.Select;
```

```
public class AccountPage {
```

```
    WebDriver driver;
```

```
    public AccountPage(WebDriver driver) {
```

```
        this.driver = driver;
```

```
    }
```

```
    By newAccount = By.linkText("New Account");
```

```
    By customerId = By.name("cusid");
```

```
    By accountType = By.name("selaccount");
```

```
    By initialDeposit = By.name("inideposit");
```



```
By submit = By.name("button2");
```

```
public void clickNewAccount() {  
    driver.findElement(newAccount).click();  
}
```

```
public void enterCustomerId(String cid) {  
    driver.findElement(customerId).sendKeys(cid);  
}
```

```
public void selectAccountType(String type) {  
    Select s = new Select(driver.findElement(accountType));  
    s.selectByVisibleText(type);  
}
```

```
public void enterInitialDeposit(String amount) {  
    driver.findElement(initialDeposit).sendKeys(amount);  
}
```

```
public void clickSubmit() {  
    driver.findElement(submit).click();  
}
```

```
}
```

Package Name – pages

Class Name- CustomerPage.java

```
package pages;
```

```
import org.openqa.selenium.By;
```

```
import org.openqa.selenium.WebDriver;
```

```
public class CustomerPage {

    WebDriver driver;

    public CustomerPage(WebDriver driver) {
        this.driver = driver;
    }

    By newCustomer = By.linkText("New Customer");
    By customerName = By.name("name");
    By gender = By.xpath("//input[@value='m']");
    By dob = By.id("dob");
    By address = By.name("addr");
    By city = By.name("city");
    By state = By.name("state");
    By pin = By.name("pinno");
    By mobile = By.name("telephoneno");
    By email = By.name("emailid");
    By password = By.name("password");
    By submit = By.name("sub");

    public void clickNewCustomer() {

        driver.findElement(newCustomer).click(); }

    public void enterCustomerName(String cname) {
        driver.findElement(customerName).sendKeys(cname);
    }
}
```

```
public void selectGender() {
```

```
    driver.findElement(gender).click();
```

```
}
```

```
public void enterDOB(String date) {
```

```
    driver.findElement(dob).sendKeys(date);
```

```
}
```

```
public void enterAddress(String addr) {
```

```
    driver.findElement(address).sendKeys(addr); }
```

```
public void enterCity(String cityname) {
```

```
    driver.findElement(city).sendKeys(cityname);
```

```
}
```

```
public void enterState(String statename) {
```

```
    driver.findElement(state).sendKeys(statename);
```

```
}
```

```
public void enterPin(String pinno) {
```

```
    driver.findElement(pin).sendKeys(pinno);
```

```
}
```

```
public void enterMobile(String mob) {
```

```
    driver.findElement(mobile).sendKeys(mob);
```

```
}
```

```
public void enterEmail(String mail) {
```

```
        driver.findElement(email).sendKeys(mail);  
    }  
}
```

```
public void enterPassword(String pass) {  
    driver.findElement(password).sendKeys(pass);  
}
```

```
public void clickSubmit() {  
    driver.findElement(submit).click();  
}
```

```
}
```

Package Name – pages

Class Name- DepositPage.java

```
package pages;
```

```
import org.openqa.selenium.By;
```

```
import org.openqa.selenium.WebDriver;
```

```
public class DepositPage {
```

```
    WebDriver driver;
```

```
    public DepositPage(WebDriver driver) {
```

```
        this.driver = driver;
```

```
    }
```

```
    By depositLink = By.linkText("Deposit");
```

```
    By accountNo = By.name("accountno");
```

```
By amount = By.name("ammount");
```

```
By description = By.name("desc");
```

```
By submitBtn = By.name("AccSubmit");
```

```
public void clickDeposit() {
```

```
    driver.findElement(depositLink).click();
```

```
}
```

```
public void enterAccountNo(String accNo) {
```

```
    driver.findElement(accountNo).sendKeys(accNo);
```

```
}
```

```
public void enterAmount(String amt) {
```

```
    driver.findElement(amount).sendKeys(amt);
```

```
}
```

```
public void enterDescription(String desc) {
```

```
    driver.findElement(description).sendKeys(desc);
```

```
}
```

```
public void clickSubmit() {
```

```
    driver.findElement(submitBtn).click();
```

```
}
```

```
}
```

Package Name – pages

Class Name- LoginPage.java

```
package pages;
```

```
import org.openqa.selenium.*;
```

```
public class LoginPage {
```

```
    WebDriver driver;
```

```
    public LoginPage(WebDriver driver) {
```

```
        this.driver = driver;
```

```
    }
```

```
    By uid =By.name("uid");
```

```
    By pass =By.name("password");
```

```
    By login =By.name("btnLogin");
```

```
    public void enterUser(String user) {
```

```
        driver.findElement(uid).sendKeys(user);
```

```
    }
```

```
    public void enterPass(String password) {
```

```
        driver.findElement(pass).sendKeys(password);
```

```
    }
```

```
    public void clickLogin() {
```

```
        driver.findElement(login).click();
```

```
    }
```

```
}
```

Package Name – pages

Class Name- WithdrawPage.java

```
package pages;
```

```
import java.time.Duration;

import org.openqa.selenium.By;
import org.openqa.selenium.JavascriptExecutor;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
```

```
public class WithdrawPage {
```

```
    WebDriver driver;
```

```
    public WithdrawPage(WebDriver driver) {
        this.driver = driver;
    }
```

```
    By withdrawLink = By.linkText("Withdrawal");
```

```
    By accountNo = By.name("accountno");
```

```
    By amount = By.name("ammount");
```

```
    By description = By.name("desc");
```

```
    By submitBtn = By.name("AccSubmit");
```

```
    public void clickWithdrawal() {
```

```
        WebDriverWait wait =
```

```
            new WebDriverWait(
```

```
                driver,
```

```
                Duration.ofSeconds(10));
```

```
WebElement ele =  
    wait.until(  
        ExpectedConditions  
        .presenceOfElementLocated(  
            By.linkText("Withdrawal")));
```

```
JavascriptExecutor js =  
    (JavascriptExecutor) driver;
```

```
js.executeScript(  
    "arguments[0].click();",  
    ele);
```

```
}
```

```
public void enterAccountNo(String accNo) {  
    driver.findElement(accountNo).sendKeys(accNo);  
}
```

```
public void enterAmount(String amt) {  
    driver.findElement(amount).sendKeys(amt);  
}
```

```
public void enterDescription(String desc) {  
    driver.findElement(description).sendKeys(desc);  
}
```

```
public void clickSubmit() {  
    driver.findElement(submitBtn).click();
```



```
}  
}
```

Package Name – pages

Class Name- FundTransferPage.java

```
package pages;
```

```
import java.time.Duration;
```

```
import org.openqa.selenium.By;
```

```
import org.openqa.selenium.JavascriptExecutor;
```

```
import org.openqa.selenium.WebDriver;
```

```
import org.openqa.selenium.WebElement;
```

```
import org.openqa.selenium.support.ui.ExpectedConditions;
```

```
import org.openqa.selenium.support.ui.WebDriverWait;
```

```
public class FundTransferPage {
```

```
    WebDriver driver;
```

```
    public FundTransferPage(WebDriver driver) {
```

```
        this.driver = driver;
```

```
    }
```

```
    By fundTransfer = By.linkText("Fund Transfer");
```

```
    By payerAccount = By.name("payersaccount");
```

```
    By payeeAccount = By.name("payeeaccount");
```

```
    By amount = By.name("ammount");
```

```
    By description = By.name("desc");
```

```
    By submitBtn = By.name("AccSubmit");
```

```
public void clickFundTransfer() {  
    System.out.println(  
        driver.findElement(fundTransfer).isDisplayed());  
    WebElement element =  
        driver.findElement(fundTransfer);  
  
    JavascriptExecutor js =(JavascriptExecutor) driver;  
    js.executeScript("arguments[0].scrollIntoView(true);",element);  
    js.executeScript("arguments[0].click();",element);  
}
```

```
public void enterPayerAccount(String payer) {  
    driver.findElement(payerAccount).sendKeys(payer);  
}
```

```
public void enterPayeeAccount(String payee) {  
    driver.findElement(payeeAccount).sendKeys(payee);  
}
```

```
public void enterAmount(String amt) {  
    driver.findElement(amount).sendKeys(amt);  
}
```

```
public void enterDescription(String desc) {  
    driver.findElement(description).sendKeys(desc);  
}
```

```
public void clickSubmit() {
```

```
        driver.findElement(submitBtn).click();  
    }  
}
```

Package Name – pages

Class Name- LogoutPage.java

```
package pages;
```

```
import java.time.Duration;
```

```
import org.openqa.selenium.By;
```

```
import org.openqa.selenium.JavascriptExecutor;
```

```
import org.openqa.selenium.WebDriver;
```

```
import org.openqa.selenium.WebElement;
```

```
import org.openqa.selenium.support.ui.ExpectedConditions;
```

```
import org.openqa.selenium.support.ui.WebDriverWait;
```

```
public class LogoutPage {
```

```
    WebDriver driver;
```

```
    By logout = By.linkText("Log out");
```

```
    public LogoutPage(WebDriver driver) {
```

```
        this.driver = driver;
```

```
    }
```

```
    public void clickLogout() {
```

```
WebDriverWait wait =  
    new WebDriverWait(  
        driver,  
        Duration.ofSeconds(10));
```

```
WebElement ele =  
    wait.until(  
        ExpectedConditions  
        .elementToBeClickable(  
            logout));
```

```
JavascriptExecutor js =  
    (JavascriptExecutor) driver;
```

```
js.executeScript("arguments[0].click();",ele);  
}  
}
```

Package Name – reports

Class Name- ExtentManager.java

```
package reports;
```

```
import com.aventstack.extentreports.ExtentReports;
```

```
import com.aventstack.extentreports.reporter.ExtentSparkReporter;
```

```
public class ExtentManager {
```

```
    public static ExtentReports getReport() {
```

```
ExtentSparkReporter reporter =  
    new ExtentSparkReporter(  
        "./Reports/ExtentReport.html");  
  
ExtentReports extent =  
    new ExtentReports();  
  
extent.attachReporter(reporter);  
  
extent.setSystemInfo(  
    "Tester",  
    "Akash");  
  
return extent;  
}  
}
```

Package Name – testCases

Class Name- LoginTest.java

```
package testcases;  
  
import org.testng.annotations.Test;  
  
import base.BaseTest;  
import pages.LoginPage;  
  
public class LoginTest extends BaseTest {  
  
    @Test(priority = 3)  
    public void validLogin() throws Exception {
```

```
login();
```

```
soft.assertTrue(  
    driver.getTitle()  
        .contains("Guru99"));
```

```
System.out.println(  
    "Valid Login Passed");  
}
```

```
@Test(priority = 2)  
public void invalidLogin() {
```

```
    LoginPage lp =new LoginPage(driver);
```

```
    lp.enterUser("ahldhld");
```

```
    lp.enterPass("1389890");
```

```
    lp.clickLogin();
```

```
    try {
```

```
        String alertText =driver.switchTo().alert()  
            .getText();
```

```
        System.out.println(alertText);
```

```
        soft.assertTrue(  
            alertText.length() > 0);
```

```
driver.switchTo().alert()
```

```
.accept();
```

```
} catch(Exception e) {
```

```
soft.assertTrue(true);
```

```
System.out.println(
```

```
"Alert not displayed");
```

```
}
```

```
System.out.println(
```

```
"Invalid Login Passed");
```

```
}
```

```
@Test(priority = 1)
```

```
public void blankLogin() {
```

```
LoginPage lp =
```

```
new LoginPage(driver);
```

```
lp.clickLogin();
```

```
try {
```

```
String msg =driver.switchTo().alert()
```

```
.getText();
```

```
System.out.println(msg);
```

```
soft.assertTrue(msg.length() > 0);
```

```
driver.switchTo().alert()
```

```
.accept();
```

```
} catch(Exception e) {
```

```
soft.assertTrue(true);
```

```
System.out.println("No Alert Displayed");
```

```
}
```

```
System.out.println("Blank Login Passed");
```

```
}
```

```
}
```

Package Name – testCases

Class Name- AccountTest.java

```
package testcases;
```

```
import org.testng.annotations.Test;
```

```
import base.BaseTest;
```

```
import pages.AccountPage;
```

```
public class AccountTest extends BaseTest {
```

```
@Test
```



```

public void openAccount() throws Exception {

    login();

    AccountPage ap =
        new AccountPage(driver);

    ap.clickNewAccount();
    ap.enterCustomerId("80841");
    ap.selectAccountType("Savings");
    ap.enterInitialDeposit("5000");
    ap.clickSubmit();

    try {

        if(driver.getPageSource()
            .contains("Account")) {

            System.out.println(
                "Account Created Successfully");
        }

    } catch(Exception e) {

        System.out.println("Account Validation Failed");
    }
}
}

```

Package Name – testCases

Class Name- CustomerTest.java

package testcases;

import org.testng.annotations.Test;

import base.BaseTest;

import pages.CustomerPage;

import utilities.CustomerData;

public class CustomerTest extends BaseTest {

**@Test(dataProvider = "customerData",
dataProviderClass = CustomerData.class)**

**public void addCustomer(
String name,
String dob,
String address,
String city,
String state,
String pin,
String mobile,
String password) throws Exception {**

login();

CustomerPage cp =new CustomerPage(driver);

cp.clickNewCustomer();

```

    cp.enterCustomerName(name);
    cp.selectGender();
    cp.enterDOB(dob);
    cp.enterAddress(address);
    cp.enterCity(city);
    cp.enterState(state);
    cp.enterPin(pin);
    cp.enterMobile(mobile);

    int random =(int)(Math.random() * 10000);

    cp.enterEmail("akash" + random + "@gmail.com");

    cp.enterPassword(password);

    cp.clickSubmit();

    soft.assertTrue(driver.getPageSource().contains("Customer"));

    System.out.println("Customer Added Successfully");
}
}

```

Package Name – testCases

Class Name- DepositTest.java

```
package testcases;
```

```
import org.testng.Assert;
```

```
import org.testng.annotations.Test;
```

```
import org.testng.asserts.SoftAssert;

import base.BaseTest;

import pages.DepositPage;

public class DepositTest extends BaseTest {

    @Test(priority = 1)

    public void validDeposit() throws Exception {

        login();

        DepositPage dp = new DepositPage(driver);

        dp.clickDeposit();
        dp.enterAccountNo("183286");
        dp.enterAmount("5000");
        dp.enterDescription("Cash Deposit");
        dp.clickSubmit();

        if(driver.getPageSource().contains("HTTP ERROR 500")) {

            System.out.println("Guru99 Deposit error Found");

        }

        soft.assertTrue(true,"Known Guru99 Defect");

        System.out.println("Deposit Successful");
    }
}
```

```
@Test(priority = 2)

public void invalidAccountDeposit() throws Exception {

    login();

    DepositPage dp = new DepositPage(driver);

    dp.clickDeposit();
    dp.enterAccountNo("999999");
    dp.enterAmount("5000");
    dp.enterDescription("Cash Deposit");
    dp.clickSubmit();

    try {

        String msg = driver.switchTo().alert().getText();
        System.out.println(msg);
        soft.assertTrue(msg.length() > 0);
        driver.switchTo().alert().accept();
        System.out.println("Invalid Account Validation Passed");

    } catch(Exception e) {

        System.out.println("Alert not displayed");

        soft.assertTrue(true, "Alert not displayed");
    }
}
```

```
        soft.assertAll();  
    }  
  
}
```

Package Name – testCases

Class Name- WithdrawTest.java

```
package testcases;
```

```
import org.testng.annotations.Test;
```

```
import base.BaseTest;
```

```
import pages.WithdrawPage;
```

```
public class WithdrawTest extends BaseTest {
```

```
    @Test(priority = 1)
```

```
    public void validWithdraw() throws Exception {
```

```
        login();
```

```
        WithdrawPage wp =
```

```
            new WithdrawPage(driver);
```

```
        wp.clickWithdrawal();
```

```
        wp.enterAccountNo("184009");
```

```
        wp.enterAmount("500");
```

```
        wp.enterDescription("ATM Withdraw");
```

```
        wp.clickSubmit();
```

```
soft.assertTrue(  
    driver.getPageSource()  
        .contains("Transaction"));  
  
System.out.println(  
    "Withdrawal Successful");  
}  
  
@Test(priority = 2)  
public void insufficientBalance() throws Exception {  
  
    login();  
  
    WithdrawPage wp =  
        new WithdrawPage(driver);  
  
    wp.clickWithdrawal();  
    wp.enterAccountNo("184009");  
    wp.enterAmount("999999");  
    wp.enterDescription("ATM Withdraw");  
    wp.clickSubmit();  
  
    try {  
  
        String msg =driver.switchTo().alert().getText();  
  
        System.out.println(msg);  
    }  
}
```

```
soft.assertTrue(  
    msg.length() > 0);
```

```
driver.switchTo()  
    .alert()  
    .accept();
```

```
} catch(Exception e) {
```

```
soft.assertTrue(true);
```

```
System.out.println(  
    "Alert not displayed");
```

```
}
```

```
System.out.println(  
    "Insufficient Balance Validation Passed");
```

```
}
```

```
}
```

Package Name – testCases

Class Name- FundTransferTest.java

```
package testcases;
```

```
import org.testng.annotations.Test;
```

```
import base.BaseTest;
```

```
import pages.FundTransferPage;
```

```
public class FundTransferTest extends BaseTest {
```


@Test(priority = 1)

public void validFundTransfer() throws Exception {

login();

FundTransferPage fp =

new FundTransferPage(driver);

fp.clickFundTransfer();

fp.enterPayerAccount("183433");

fp.enterPayeeAccount("183286");

fp.enterAmount("200");

fp.enterDescription("Fund Transfer");

fp.clickSubmit();

soft.assertTrue(

driver.getPageSource()

.contains("Fund"));

System.out.println(

"Fund Transfer Successful");

}

@Test(priority = 2)

public void invalidAccountTransfer() throws Exception {

login();

```
FundTransferPage fp =  
    new FundTransferPage(driver);  
  
fp.clickFundTransfer();  
fp.enterPayerAccount("183433");  
fp.enterPayeeAccount("999999");  
fp.enterAmount("1000");  
fp.enterDescription("Fund Transfer");  
fp.clickSubmit();  
  
try {  
  
    String msg = driver.switchTo().alert().getText();  
  
    System.out.println(msg);  
  
    soft.assertTrue(  
        msg.length() > 0);  
  
    driver.switchTo().alert().accept();  
  
} catch (Exception e) {  
  
    soft.assertTrue(true);  
  
    System.out.println(  
        "Alert not displayed");  
}
```

```
System.out.println(  
    "Invalid Account Validation Passed");  
}
```

```
@Test(priority = 3)  
public void insufficientBalanceTransfer() throws Exception {
```

```
    login();
```

```
    FundTransferPage fp =  
        new FundTransferPage(driver);
```

```
    fp.clickFundTransfer();  
    fp.enterPayerAccount("183433");  
    fp.enterPayeeAccount("183286");  
    fp.enterAmount("9999999");  
    fp.enterDescription("Fund Transfer");  
    fp.clickSubmit();
```

```
    try {
```

```
        String msg =driver.switchTo().alert()  
            .getText();
```

```
        System.out.println(msg);
```

```
        soft.assertTrue(  
            msg.length() > 0);
```

```

        driver.switchTo().alert()
            .accept();

    } catch (Exception e) {

        soft.assertTrue(true);

        System.out.println(
            "Alert not displayed");
    }

    System.out.println(
        "Insufficient Balance Validation Passed");
    }
}

```

Package Name – testCases

Class Name- LogoutTest.java

```
package testcases;
```

```
import org.openqa.selenium.Alert;
```

```
import org.testng.Assert;
```

```
import org.testng.annotations.Test;
```

```
import base.BaseTest;
```

```
import pages.LogoutPage;
```

```
public class LogoutTest extends BaseTest {
```

```
    @Test
```

```
public void validLogout() throws Exception {

    login();

    LogoutPage lp = new LogoutPage(driver);

    lp.clickLogout();

    Alert alert = driver.switchTo().alert();

    String msg = alert.getText();

    Assert.assertEquals(msg, "You Have Succesfully Logged Out!!");

    alert.accept();

    System.out.println("Logout Successful");
}
}
```

Package Name – Utilities

Class Name- ConfigReader.java

```
package utilities;

import java.io.FileInputStream;
import java.util.Properties;

public class ConfigReader {
```

Properties prop;

public ConfigReader() throws Exception {

FileInputStream fis =

new FileInputStream(

"src/test/resources/config.properties");

prop = new Properties();

prop.load(fis);

}

public String getBrowser() {

return prop.getProperty("browser");

}

public String getUrl() {

return prop.getProperty("url");

}

}

Package Name – Utilities

Class Name- CustomerData.java

package utilities;

import org.testng.annotations.DataProvider;

```

public class CustomerData {

    @DataProvider(name = "customerData")
    public Object[][] getCustomerData() {

        return new Object[][] {

            {"Akashsingh","01052002","Kolkata","Kolkata",
            "West Bengal","700001","9876543210","abc123"}

        };

    }
}

```

Package Name – Utilities

Class Name- ExcelUtils.java

```

package utilities;

import java.io.FileInputStream;

import org.apache.poi.xssf.usermodel.*;

public class ExcelUtils {

    public static String getData(String sheet,int row,int cell)throws Exception {

        FileInputStream fis =new FileInputStream("project.xlsx");
        XSSFWorkbook wb =new XSSFWorkbook(fis);
        XSSFSheet sh = wb.getSheet(sheet);
        return sh.getRow(row).getCell(cell).getStringCellValue();

    }
}

```

```
}
```

Package Name – Utilities

Class Name- ScreenshotUtil.java

```
package utilities;
```

```
import java.io.File;
```

```
import java.io.IOException;
```

```
import org.apache.commons.io.FileUtils;
```

```
import org.openqa.selenium.OutputType;
```

```
import org.openqa.selenium.TakesScreenshot;
```

```
import org.openqa.selenium.WebDriver;
```

```
public class ScreenshotUtil {
```

```
    public static void captureScreenshot(WebDriver driver,
```

```
        String testName)
```

```
        throws IOException {
```

```
        File src = ((TakesScreenshot) driver)
```

```
            .getScreenshotAs(OutputType.FILE);
```

```
        File dest = new File(
```

```
            "./Screenshots/"
```

```
            + testName + ".png");
```

```
        FileUtils.copyFile(src, dest);
```

```
    }
```



```
}
```

Package Name – Utilities

Class Name- WaitUtils.java

```
package utilities;
```

```
import java.time.Duration;
```

```
import org.openqa.selenium.By;
```

```
import org.openqa.selenium.WebDriver;
```

```
import org.openqa.selenium.WebElement;
```

```
import org.openqa.selenium.support.ui.ExpectedConditions;
```

```
import org.openqa.selenium.support.ui.WebDriverWait;
```

```
public class WaitUtils {
```

```
    public static WebElement waitForElement(
```

```
        WebDriver driver,
```

```
        By locator) {
```

```
        WebDriverWait wait =
```

```
            new WebDriverWait(
```

```
                driver,
```

```
                Duration.ofSeconds(10));
```

```
        return wait.until(
```

```
            ExpectedConditions
```

```
                .visibilityOfElementLocated(locator));
```

```
    }
```

```
public static void waitForClick(  
    WebDriver driver,  
    By locator) {  
  
    WebDriverWait wait =  
        new WebDriverWait(  
            driver,  
            Duration.ofSeconds(10));  
  
    wait.until(  
        ExpectedConditions  
            .elementToBeClickable(locator));  
    }  
}
```

JenkinsFile

```
pipeline {  
    agent any  
  
    tools {  
        maven 'Maven3'  
        jdk 'JDK'  
    }  
  
    stages {  
  
        stage('Checkout') {  
            steps {  
                git branch: 'main',
```

```
url: 'https://github.com/Akashraj44/capstone_project.git'
```

```
}
```

```
}
```

```
stage('Clean') {
```

```
  steps {
```

```
    dir('Capstone_Project') {
```

```
      bat 'mvn clean'
```

```
    }
```

```
  }
```

```
}
```

```
stage('Compile') {
```

```
  steps {
```

```
    dir('Capstone_Project') {
```

```
      bat 'mvn compile'
```

```
    }
```

```
  }
```

```
}
```

```
stage('Test') {
```

```
  steps {
```

```
    dir('Capstone_Project') {
```

```
      bat 'mvn test'
```

```
    }
```

```
  }
```

```
}
```

```
stage('Package') {
```

```
steps {  
  dir('Capstone_Project') {  
    bat 'mvn package'  
  }  
}  
}
```

```
stage('Docker Build') {  
  steps {  
    dir('Capstone_Project') {  
      bat 'docker build -t guru99-framework .'  
    }  
  }  
}
```

```
stage('Docker Run') {  
  steps {  
    dir('Capstone_Project') {  
      bat 'docker run --rm guru99-framework'  
    }  
  }  
}
```

```
post {  
  success {  
    echo 'Guru99 Framework Build Successful'  
  }  
}
```

```
failure {  
    echo 'Guru99 Framework Build Failed'  
}
```

```
always {  
    echo 'Pipeline Execution Completed'  
}
```

}

}

 Jenkins

guru99_project

#12

Stages

Started by AKASH RAJ

Started 8 min 24 sec ago

Queued 0.31 sec

Build has been executing for 8 min 24 sec

Changes

Start

✓

Checkout SCM

✓

Tool Install

✓

Checkout

✓

Clean

✓

Compile

✓

Test

✓

Package

✓

Post Actions

End

+

-

↺

Q Search

≡

✓ Checkout SCM 5s

✓ Tool Install 0.22s

✓ Checkout 2s

✓ Clean 6s

✓ Compile 16s

✓ Test 5m 18s

✓ Package 4m 54s

✓ Post Actions 0.16s

✓ Post Actions 0.16s Started 4s ago Jenkins

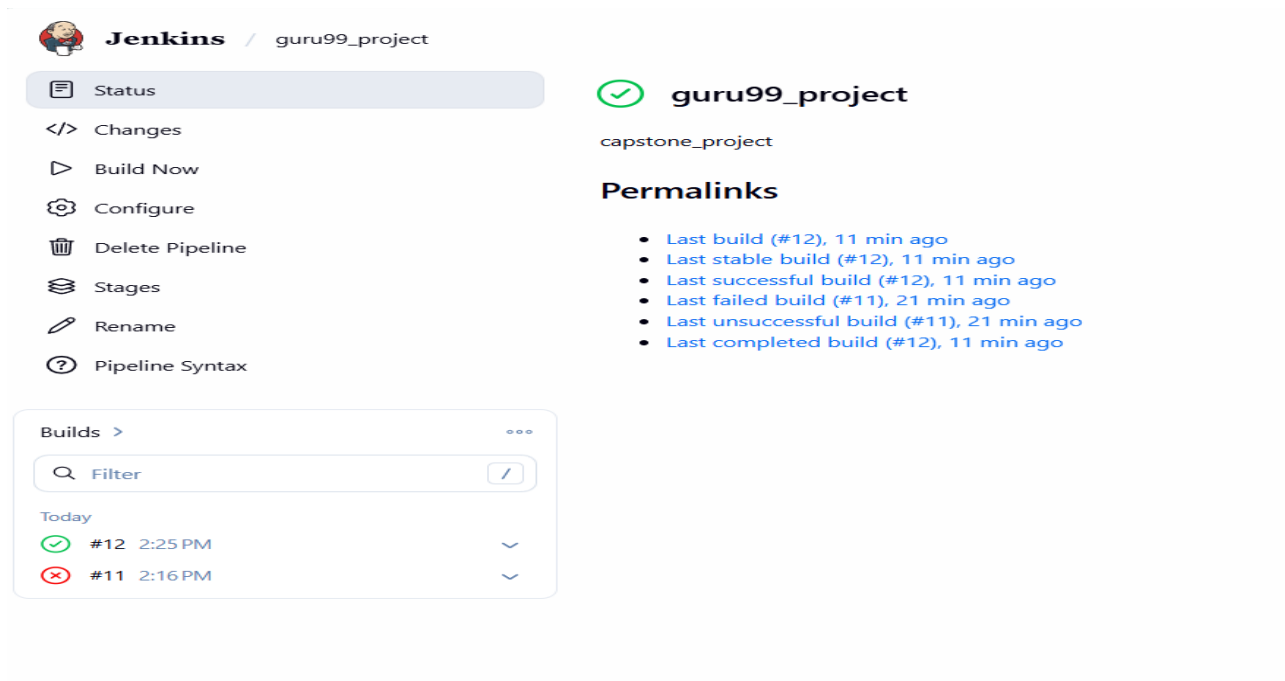
✓ Pipeline Execution Completed 53ms

✓ Guru99 Framework Build Successful 42ms

Guru99 Framework Build Successful

↑

↓



Jenkins / guru99_project

Status

- Changes
- Build Now
- Configure
- Delete Pipeline
- Stages
- Rename
- Pipeline Syntax

guru99_project
capstone_project

Permalinks

- Last build (#12), 11 min ago
- Last stable build (#12), 11 min ago
- Last successful build (#12), 11 min ago
- Last failed build (#11), 21 min ago
- Last unsuccessful build (#11), 21 min ago
- Last completed build (#12), 11 min ago

Builds

Filter

Today

- #12 2:25 PM
- #11 2:16 PM

Dockerfile

FROM maven:3.9.11-eclipse-temurin-21

Install dependencies

RUN apt-get update && apt-get install -y \

wget \

gnupg \

unzip \

curl

Add Google Chrome repository

RUN wget -q -O - https://dl.google.com/linux/linux_signing_key.pub | \

gpg --dearmor -o /usr/share/keyrings/google-chrome.gpg

RUN echo "deb [arch=amd64 signed-by=/usr/share/keyrings/google-chrome.gpg] \

http://dl.google.com/linux/chrome/deb/ stable main" \

> /etc/apt/sources.list.d/google-chrome.list

Install Chrome

RUN apt-get update && apt-get install -y google-chrome-stable

WORKDIR /app

COPY . .

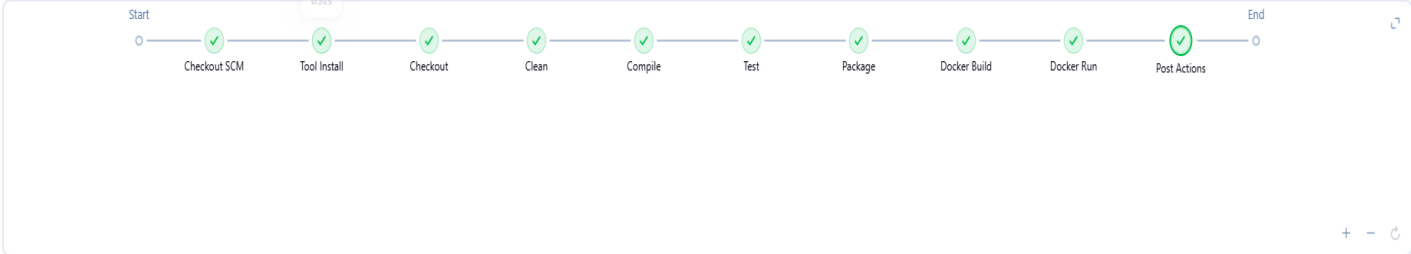
RUN mvn clean install -DskipTests

CMD ["mvn","test"]

 **Jenkins** / guru99_project_doc / #6 / Stages

 < #6  

 Started by AKASH RAJ  Started 22 min ago  Queued 38 ms  Took 17 min • </> Changes



 Search 

 Checkout SCM 4s

 Tool Install 0.32s

 Checkout 3s

 Clean 13s

 Compile 19s

 Test 5m 37s

 Package 4m 30s

 Docker Build 2m 44s

 Docker Run 4m 14s

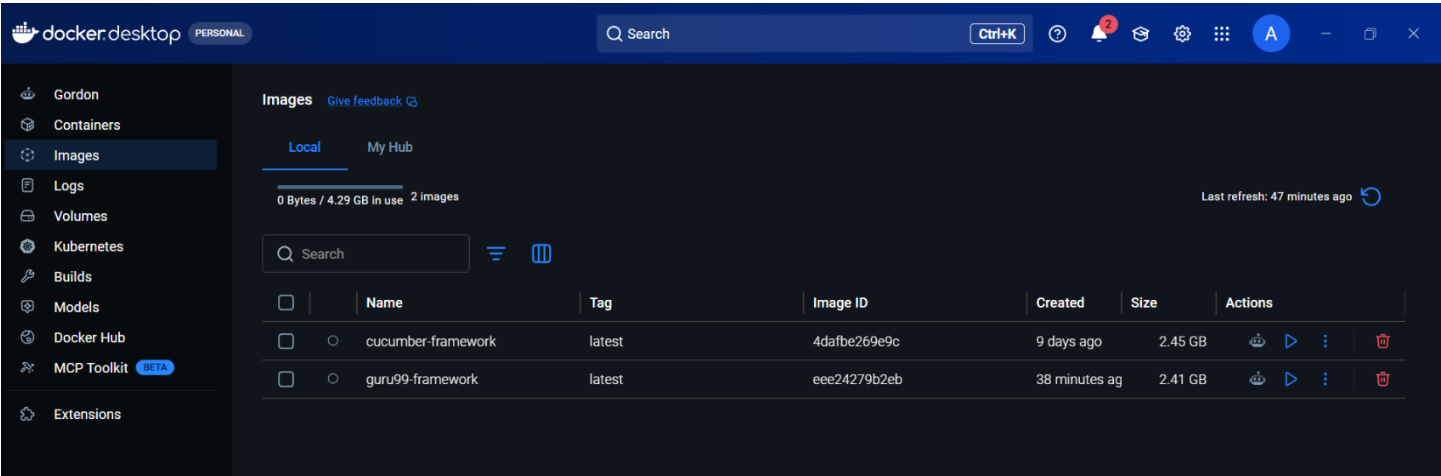
 Post Actions 0.18s

 Post Actions  0.18s  Started 5m ago  Jenkins 

 Pipeline Execution Completed  57ms 

 Guru99 Framework Build Successful  26ms 

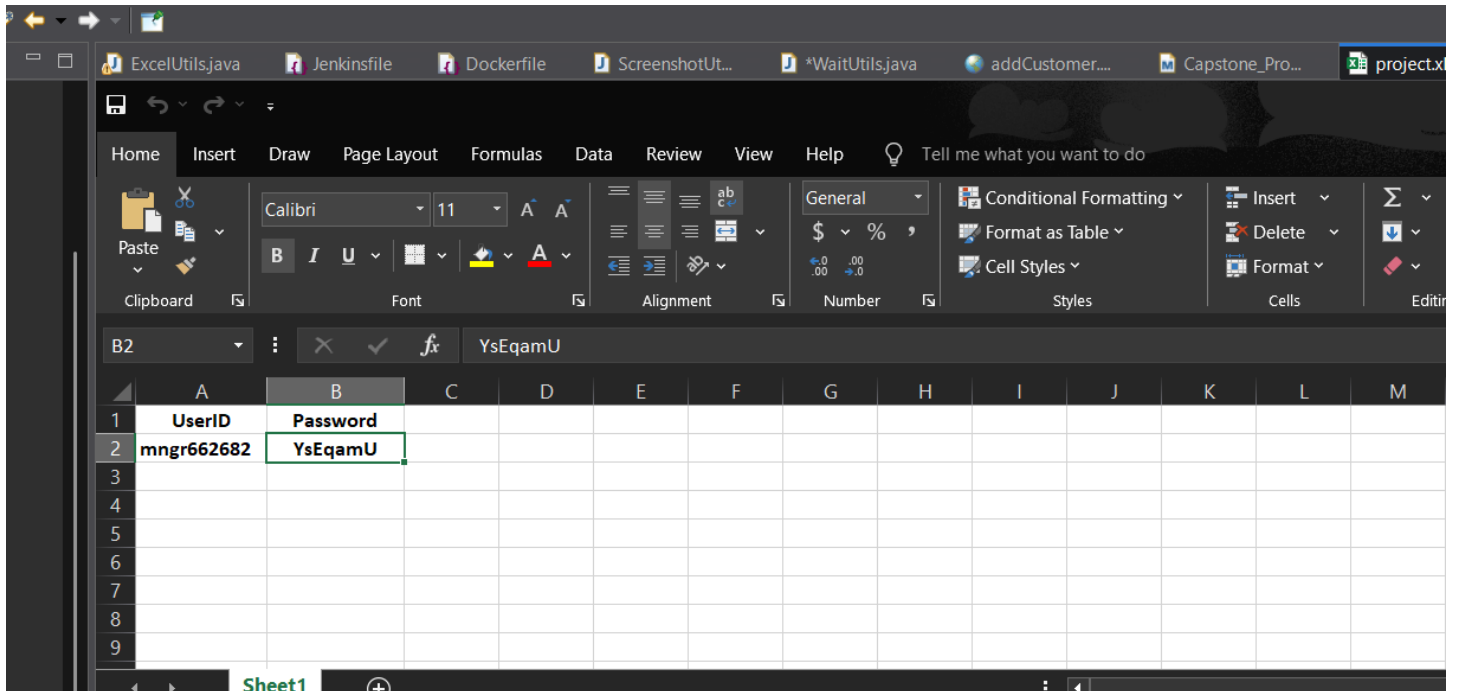




Pom.xml

```
1 https://maven.apache.org/xsd/maven-4.0.0.xsd (xsi:schemaLocation)
2 <?xml version="1.0" encoding="UTF-8" ?>
3 <project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 https://maven.apache.org/xsd/maven-4.0.0.xsd">
4   <modelVersion>4.0.0</modelVersion>
5   <groupId>Capstone_Project</groupId>
6   <artifactId>Capstone_Project</artifactId>
7   <version>0.0.1-SNAPSHOT</version>
8   <dependencies>
9     <!-- Source: https://mvnrepository.com/artifact/org.testng/testng -->
10    <dependency>
11      <groupId>org.testng</groupId>
12      <artifactId>testng</artifactId>
13      <version>7.12.0</version>
14      <scope>test</scope>
15    </dependency>
16    <!-- Source: https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
17    <dependency>
18      <groupId>org.seleniumhq.selenium</groupId>
19      <artifactId>selenium-java</artifactId>
20      <version>4.44.0</version>
21      <scope>compile</scope>
22    </dependency>
23    <dependency>
24      <groupId>org.apache.poi</groupId>
25      <artifactId>poi-ooxml</artifactId>
26      <version>5.2.5</version>
27    </dependency>
28    <dependency>
29      <groupId>commons-io</groupId>
30      <artifactId>commons-io</artifactId>
31      <version>2.16.1</version>
32    </dependency>
33    <dependency>
34      <groupId>com.aventstack</groupId>
35      <artifactId>extentreports</artifactId>
36      <version>5.1.1</version>
37    </dependency>
38  </dependencies>
39  <build>
40    <plugins>
```


Project.xlsx



The screenshot shows a Microsoft Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	UserID	Password											
2	mngnr662682	YsEqamU											
3													
4													
5													
6													
7													
8													
9													

The spreadsheet is titled "Sheet1" and the active cell is B2, containing the text "YsEqamU". The ribbon shows the "Home" tab with various formatting options.

Testing.xml

```
https://testng.org/testng-1.0.dtd (doctype)
1  <?xml version="1.0" encoding="UTF-8"?>
2  <!DOCTYPE suite SYSTEM "https://testng.org/testng-1.0.dtd">
3  <suite name="Guru99 Suite">
4    <listeners>
5      <listener
6        class-name=
7          "listeners.ListenerClass"/>
8      <listener
9        class-name=
10         "listeners.ExtentListener"/>
11    </listeners>
12    <test name="Guru99 Tests">
13      <classes>
14        <class name=
15          "testcases.LoginTest"/>
16
17        <class name=
18          "testcases.CustomerTest"/>
19
20        <class name=
21          "testcases.AccountTest"/>
22
23        <class name=
24          "testcases.DepositTest"/>
25
26        <class name=
27          "testcases.WithdrawTest"/>
28
29        <class name=
30          "testcases.FundTransferTest"/>
31
32        <class name="testcases.LogoutTest"/>
33
34      </classes>
35    </test>
36  </suite>
37
38
39
40
```

Console output

Chrome browser started

Test Started : blankLogin

No Alert Displayed

Blank Login Passed

Test Passed : blankLogin

Chrome browser started

Test Started : invalidLogin

Alert not displayed

Invalid Login Passed

Test Passed : invalidLogin

Chrome browser started

Test Started : validLogin

Valid Login Passed

Test Passed : validLogin

Chrome browser started

Test Started : addCustomer

Customer Added Successfully

Test Passed : addCustomer

Chrome browser started

Test Started : openAccount

Account Created Successfully

Test Passed : openAccount

Chrome browser started

Test Started : validDeposit

Deposit Successful

Test Passed : validDeposit

Chrome browser started

Test Started : invalidAccountDeposit

Alert not displayed

Test Passed : invalidAccountDeposit

Chrome browser started

Test Started : validWithdraw

Withdrawal Successful

Test Passed : validWithdraw

Chrome browser started

Test Started : insufficientBalance

Transaction Failed. Account Balance Low!!!

Insufficient Balance Validation Passed

Test Passed : insufficientBalance

Chrome browser started

Test Started : validFundTransfer

true

Fund Transfer Successful

Test Passed : validFundTransfer

Chrome browser started

Test Started : invalidAccountTransfer

true

Account 999999does not exist!!!

Invalid Account Validation Passed

Test Passed : invalidAccountTransfer

Chrome browser started

Test Started : insufficientBalanceTransfer

true

Alert not displayed

Insufficient Balance Validation Passed

Test Passed : insufficientBalanceTransfer

Chrome browser started

Test Started : validLogout

Logout Successful

Test Passed : validLogout

=====

Guru99 Suite

Total tests run: 13, Passes: 13, Failures: 0, Skips: 0

=====

Search:

Passed: 13Failed: 0Skipped: 0

Tests: 1/1Methods: 13 (178532 ms)

All TestsFailed TestsSummary

Guru99 Suite (13/0/0/0) (65.794 s)

Guru99 Tests (65.794 s)

testcases.LoginTest

blankLogin (0.248 s)

invalidLogin (0.605 s)

validLogin (8.54 s)

testcases.CustomerTest

addCustomer (7.424 s)

"Akashsingh","01052002","Kolkata","Kolkata","West Bengal","700001","9876543210","abc123" (7.424 s)

testcases.AccountTest

openAccount (5.666 s)

testcases.DepositTest

validDeposit (3.754 s)

invalidAccountDeposit (3.857 s)

testcases.WithdrawTest

validWithdraw (5.156 s)

insufficientBalance (5.558 s)

testcases.FundTransferTest

validFundTransfer (4.064 s)

invalidAccountTransfer (5.392 s)

insufficientBalanceTransfer (11.281 s)

testcases.LogoutTest

validLogout (4.249 s)

Failure Exception

report

1 suite

Switch Retro Theme

Test results

All suites

Guru99 Suite

Info

C:\Users\Akash

raj\git\capstone_project\Capstone_Project\testing.xml

1 test

0 groups

Times

Reporter output

Ignored methods

Chronological view

Results

13 methods, 13 passed

Passed methods (hide)

addCustomer(Akashsingh, 01052002, Kolkata, Kolkata, West Bengal, 700001, 9876543210, abc123)

blankLogin

insufficientBalance

insufficientBalanceTransfer

invalidAccountDeposit

invalidAccountTransfer

invalidLogin

openAccount

validDeposit

validFundTransfer

validLogin

validLogout

validWithdraw

testcases.LogoutTest

validLogout

testcases.AccountTest

openAccount

testcases.FundTransferTest

insufficientBalanceTransfer

invalidAccountTransfer

validFundTransfer

testcases.LoginTest

blankLogin

invalidLogin

validLogin

testcases.WithdrawTest

insufficientBalance

validWithdraw

testcases.DepositTest

invalidAccountDeposit

validDeposit

testcases.CustomerTest

addCustomer (Akashsingh,01052002,Kolkata,Kolkata,West Bengal,700001,9876543210,abc123)